☑ CTL Provisional DECISION SHEET (PDSH)
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Standard(s) (incl. year)	**Subclause(s)**	**Tracking No.**
IEC 60947-1:2020	Annex N.3.2.1 – Creepages	PDSH 2264
Category		
POW		
Subject	**Keywords**	**Developed by**
Application of Annex N of IEC 60947-1:2020 for accessories with a limited voltage range	Annex N, accessories, limited voltage range	ETF 7

Question

When the connection of an equipment to a safety extra-low voltage (SELV) or protective extra-low voltage (PELV) circuit is allowed by the manufacturer only when the voltage of the main circuit of the equipment is lower than a specified value U_e (e.g. 440V), this specified value being lower than the U_i of the equipment (e.g. 630V), is it allowed to apply Annex N of IEC 60947-1:2020 with the maximum voltage U_e allowed by the manufacturer's documentation (440V in the example) instead of the U_i of the equipment (630V in the example) for the verification of creepage distances?

Refer to annex for schematic diagram.

Decision

When referring to Annex N of IEC 60947-1:2020, the following instructions apply to clause N.3.2.1:

When the connection of an equipment to a safety extra-low voltage (SELV) or protective extra-low voltage (PELV) circuit is allowed by the manufacturer, the maximum rated operational voltage (U_e) of the main circuit specified by the manufacturer for the use of these circuits shall be considered in the verification of corresponding creepages in clause N.3.2.1.

Explanatory notes

Enquiry:

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In the conditions described above, the verification of creepage distances considering the maximum allowed main circuit rated operational voltage U_e (440V in this example) in the first column of Table 15 of IEC 60947-1:2020 ensures the required insulation.

Annex N, clause N.3.2.1 does not specify the reference voltage to be considered for the verification of corresponding creepages.

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Annex to PDSH 2264

